LinearSkills Academy

Class 15 — Lambda + Built-ins

Simple English — For Everyone

1. Lambda Function

Simple Meaning: A lambda is a tiny function that has no name. It does one small job in just one line.

```
square = lambda x: x * x
print(square(5))   # 25
```

This is the same as writing a normal function, just shorter:

```
def square(x):
    return x * x
```

Use lambda only for very small tasks. For bigger tasks, use a normal def function.

2. map()

Simple Meaning: map() takes a list and does the same action to every single item in it.

```
numbers = [1, 2, 3, 4]
squared = list(map(lambda x: x * x, numbers))
print(squared)   # [1, 4, 9, 16]
```

Think of it like a machine: you put in a list, every item goes through the same action, and a new list comes out.

3. filter()

Simple Meaning: filter() checks every item in a list and only keeps the ones that pass a test (True or False).

```
numbers = [1, 2, 3, 4, 5, 6]
evens = list(filter(lambda x: x % 2 == 0, numbers))
print(evens)   # [2, 4, 6]
```

Think of it like a sieve: only the items that pass the test go through.

4. sorted()

Simple Meaning: sorted() arranges a list in order. With key, you can tell it exactly what to sort by.

```
names = ["Bilal", "Ali", "Sara"]
print(sorted(names))   # ['Ali', 'Bilal', 'Sara']
```

Sorting by a custom rule, like sorting students by their marks:

```
students = [("Ali", 85), ("Sara", 92), ("Bilal", 78)]
by_marks = sorted(students, key=lambda s: s[1])
print(by_marks)   # [('Bilal', 78), ('Ali', 85), ('Sara', 92)]
```

Add reverse=True to sort from highest to lowest instead.

5. zip()

Simple Meaning: zip() joins two lists together, matching item 1 with item 1, item 2 with item 2, and so on.

```
names = ["Ali", "Sara", "Bilal"]
marks = [85, 92, 78]
```

```
combined = list(zip(names, marks))
print(combined)
# [('Ali', 85), ('Sara', 92), ('Bilal', 78)]
```

Think of a zipper on a jacket: it pulls two sides together, one pair at a time.

Looping through zipped data:

```
for name, mark in zip(names, marks):
    print(name, "scored", mark)
```

6. Quick Reference Table

Function	Simple Meaning	Example
`lambda`	A tiny function with no name	`lambda x: x + 1`
`map()`	Does the same thing to every item	`map(func, list)`
`filter()`	Keeps only items that pass a test	`filter(func, list)`
`sorted()`	Puts a list in order	`sorted(list, key=func)`
`zip()`	Joins two lists together, pair by pair	`zip(list1, list2)`

7. Small Project — Student Score Report

This project uses everything from this class together: zip, filter, sorted, and lambda.

```
names = ["Ali", "Sara", "Bilal", "Zara"]
marks = [85, 92, 45, 78]

students = list(zip(names, marks))

# Keep only students who passed (marks >= 50)
passed = list(filter(lambda s: s[1] >= 50, students))

# Sort passed students by marks, highest first
ranked = sorted(passed, key=lambda s: s[1], reverse=True)

for name, mark in ranked:
    print(name, "-", mark)
```

Output: Sara - 92, Ali - 85, Bilal - 78 (Zara is left out because 45 is a fail).